List of some commonly used learning algorithms

- Decision trees
- Neural networks
- Support vector machines
- Random forests
- Nearest neighbour algorithms
- …

Intermezzo: the binary system

Decimal system: 5 273

10000	1000	100	10	1
0	5	2	7	3

Every column represents a decimal, and every decimal digit has a value from 0 to 9 (inclusive)

Binary system: 1 / 2 / 9 / 15

16	8	4	2	1		
0	0	0	0	1	→	1 in decimal
0	0	0	1	0	→	2 in decimal
0	1	0	0	1	→	9 in decimal
0	1	1	1	1	→	15 in decimal

Every column represents one bit, and every bit (binary digit) has a value of 0 or 1

Binary counting in the early 18th century with Leibniz (in French!)

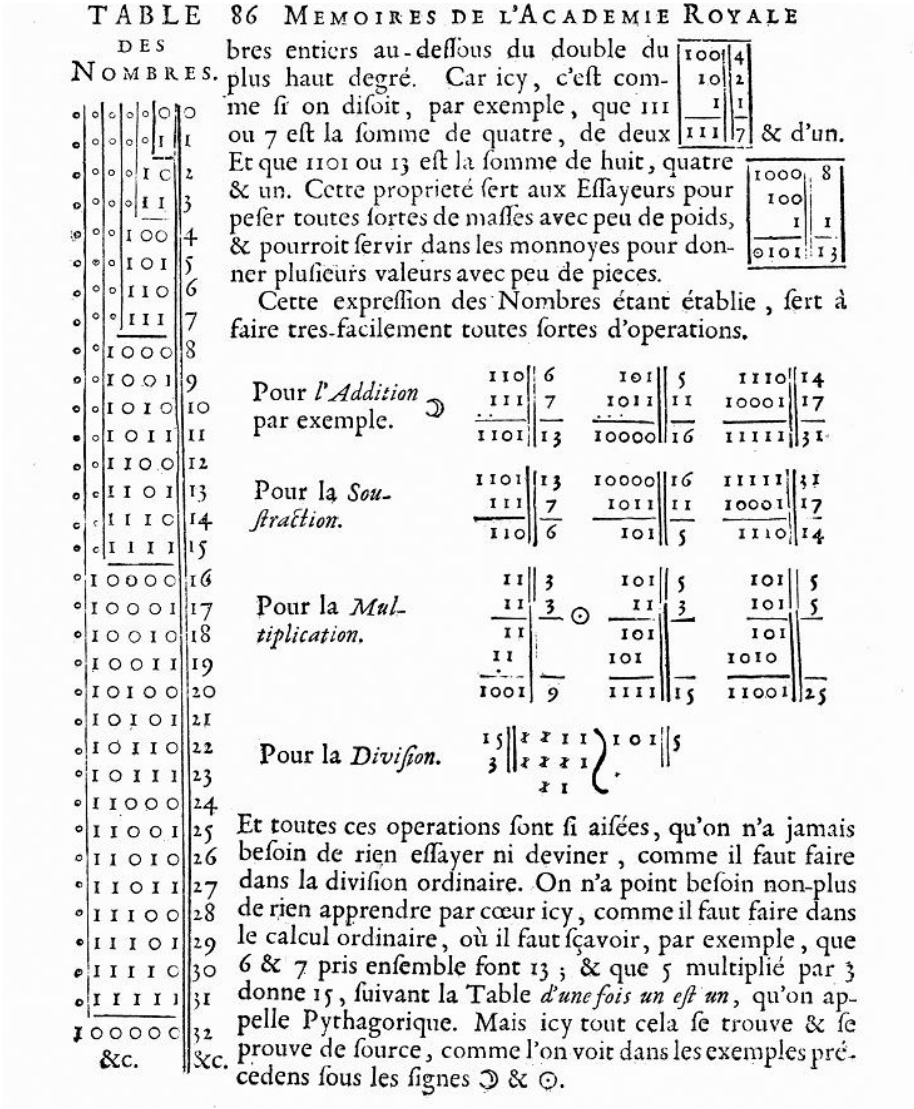


TABLE DES NOMBRES.

0	0
1	1
10	2
11	3
100	4
101	5
110	6
111	7
1000	8
1001	9
1010	10
1011	11
1100	12
1101	13
1110	14
1111	15
10000	16
10001	17
10010	18
10011	19
10100	20
10101	21
10110	22
10111	23
11000	24
11001	25
11010	26
11011	27
11100	28
11101	29
11110	30
11111	31
100000	32
&c.	&c.

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bres entiers au-dessous du double du plus haut degré. Car icy, c'est comme si on disoit, par exemple, que 111 ou 7 est la somme de quatre, de deux & d'un. Et que 1101 ou 13 est la somme de huit, quatre & un. Cette proprieté sert aux Essayeurs pour peser toutes sortes de masses avec peu de poids, & pourroit servir dans les monnoyes pour donner plusieurs valeurs avec peu de pieces.

Cette expression des Nombres étant établie, sert à faire tres-facilement toutes sortes d'operations.

Pour *l'Addition* par exemple. ☽

Pour la *Soustraction.*

Pour la *Multiplication.* ⊙

Pour la *Division.*

Et toutes ces operations sont si aisées, qu'on n'a jamais besoin de rien essayer ni deviner, comme il faut faire dans la division ordinaire. On n'a point besoin non-plus de rien apprendre par cœur icy, comme il faut faire dans le calcul ordinaire, où il faut sçavoir, par exemple, que 6 & 7 pris ensemble font 13; & que 5 multiplié par 3 donne 15, suivant la Table *d'une fois un est un*, qu'on appelle Pythagorique. Mais icy tout cela se trouve & se prouve de source, comme l'on voit dans les exemples précedens sous les signes ☽ & ⊙.

Explication de l'Arithmétique Binaire, Gottfried Wilhelm von Leibniz, 1703

Entropy, bits, and states

DNA has four bases (or four possible 'states'): A, C, G, and T

We can code every base ('state') as a number:

A → 0 C → 1 G → 2 T → 3

We can write these four decimal numbers in binary as well:

base	decimal	binary
A	0	00
C	1	01
G	2	10
T	3	11

→ We need two bits to encode these four bases.

Maximal entropy for a system

The maximal entropy of a system is equal to the minimal number of bits needed to encode all possible states of that systeem.

$$\text{maximal entropy} = \log_2(\#\text{states})$$

Logarithm base 2 because we count in bits, that can take one of two values: 0 or 1

Number of possible states that the system can occupy, e.g., four different states for a DNA base

Some examples of maximal entropy

- The maximal entropy of the biological system "sex" is 1
  - two states are possible: XY en XX
  - 1 bit is enough to code this:
    - XY → 0 and XX → 1
  - or:
    - $\log_2(2) = 1$

- The maximal entropy of a single letter in the western alphabet is just over 4.7
  - There are 26 possible states: A to Z (inclusive)
  - We need more than four bits, but less than five complete bits to encode this:
    - A → 0 / B → 1 / … / Y → 24 / Z → 25 and 25 = 11001 in binary
  - More precisely:
    - $\log_2(26) = 4.70044$ (*rounded*)

The concept of local entropy

The local entropy of a system is calculated based on the frequency with which every actually observed state occurs.

$$\text{Local entropy} = -\sum (p_i \log_2(p_i))$$

Sum across all actually occuring states

Frequency of this state

Some examples of local entropy

- De local entropy of the starting years of first-year biomedical sciences students is roughly 0.59

  There are 181 students, and 3 effective states (“2014”, “2015”, en “2016”)

  “2016”: 160/181 ; “2015”: 18/181 ; “2014”: 3/181

  – { (160/181) * $\log_2$(160/181) +

  (18/181) * $\log_2$(18/181) +

  (3/181) * $\log_2$(3/181) } = 0.586 *(rounded)*

- The local entropy of the first letters of every word in this sentence: approximately 2.9

  t, l, e, o, t, f, l, o, e, w, i, t, s → 13 letters, of which 8 are unique

  t: 3/13 ; l: 2/13 ; e: 2/13 ; o: 2/13 ; f: 1/13 ; w: 1/13 ; i: 1/13 ; s: 1/13

  – { (3/13)*$\log_2$(3/13) + (2/13)*$\log_2$(2/13) + (2/13)*$\log_2$(2/13) +

  (2/13)*$\log_2$(2/13) + (1/13)*$\log_2$(1/13)) + (1/13)*$\log_2$(1/13) +

  (1/13)*$\log_2$(1/13) + (1/13)*$\log_2$(1/13)} = 2.873 *(rounded)*

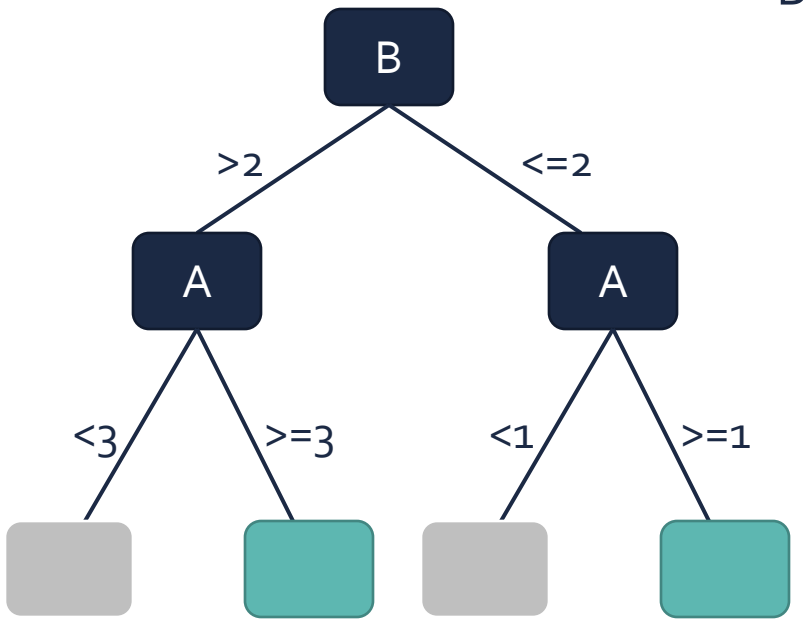
Local entropy and maximal entropy for a system

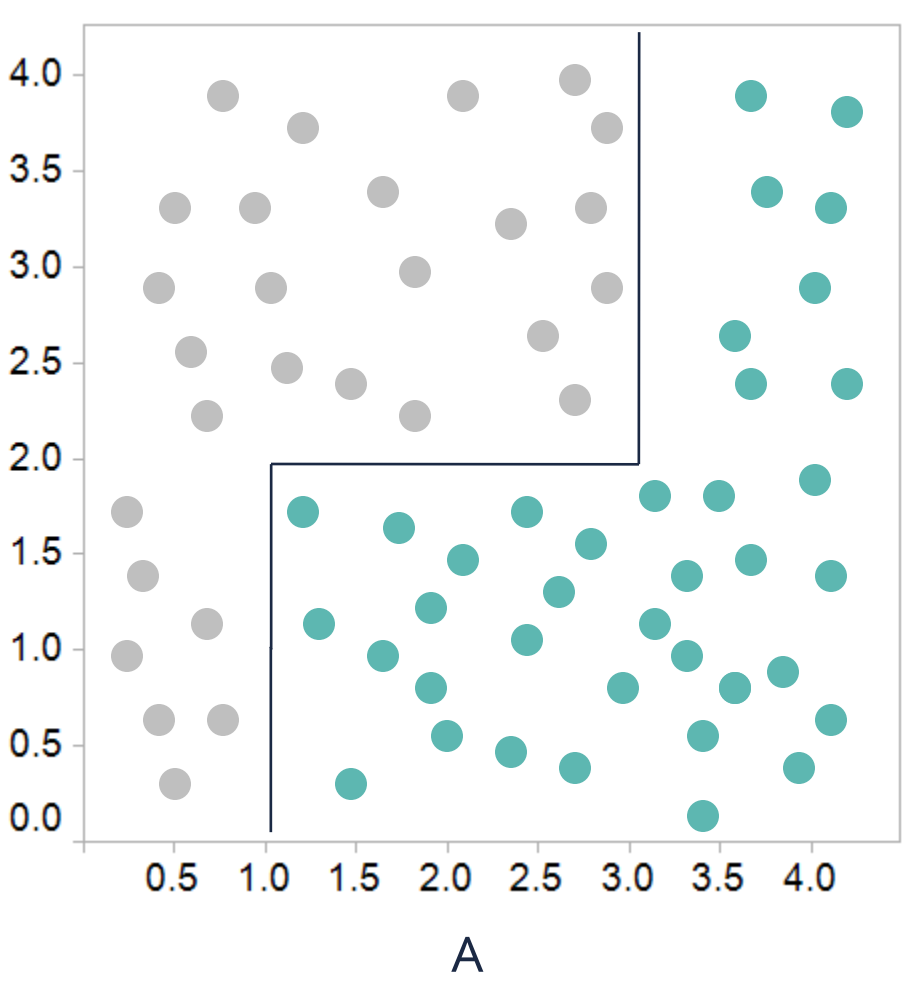
- The maximal entropy of the starting years of first-year biomedical sciences students is: $\log_2$(3 possibilities) = 1.58 (*rounded*)
- The lcoal entropy for AY 2016-2017, as calculated on the previous slide, is: 0.586 (*rounded*)
- The local entropy is about a third of the maximal entropy – indicating a relatively homogeneous composition

- Let us now change the composition of the first-year students as follows:

  "2016": 101/181 ; "2015": 46/181 ; "2014": 34/181
- The local entropy now becomes: 1.425 (*rounded*)
- The local entropy now becomes almost equal to the maximal entropy (*1.58, see top*) –indicating a quite randomly dispersed composition

Decision tree





Induction of a decision tree

```
EXAMPLES
Parameters: Culture, Fly, Hot, Music, Nature, Likes;

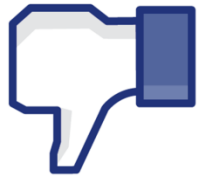
Training Examples:
1: 0, 0, 1, 0, 0, 0;
2: 0, 1, 1, 0, 0, 0;
3: 1, 1, 1, 1, 1, 0;
4: 0, 1, 1, 1, 1, 0;
5: 0, 1, 1, 0, 1, 0;
6: 1, 0, 0, 1, 1, 1;
7: 0, 0, 0, 0, 0, 0;
8: 0, 0, 0, 1, 1, 1;
9: 1, 1, 1, 0, 0, 0;
10: 1, 1, 0, 1, 1, 1;
11: 1, 1, 0, 0, 0, 1;
12: 1, 0, 1, 0, 1, 1;
13: 0, 0, 0, 1, 0, 0;
14: 1, 0, 1, 1, 0, 0;
```



decision tree



of



Induction of a decision tree

```
EXAMPLES
Parameters: Culture, Fly, Hot, Music, Nature, Likes;

Training Examples:
1: 0, 0, 1, 0, 0, 0;
2: 0, 1, 1, 0, 0, 0;
3: 1, 1, 1, 1, 1, 0;
4: 0, 1, 1, 1, 1, 0;
5: 0, 1, 1, 0, 1, 0;
6: 1, 0, 0, 1, 1, 1;
7: 0, 0, 0, 0, 0, 0;
8: 0, 0, 0, 1, 1, 1;
9: 1, 1, 1, 0, 0, 0;
10: 1, 1, 0, 1, 1, 1;
11: 1, 1, 0, 0, 0, 1;
12: 1, 0, 1, 0, 1, 1;
13: 0, 0, 0, 1, 0, 0;
14: 1, 0, 1, 1, 0, 0;
```

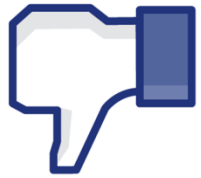
Information Entropy Minimalisation



decision tree



of

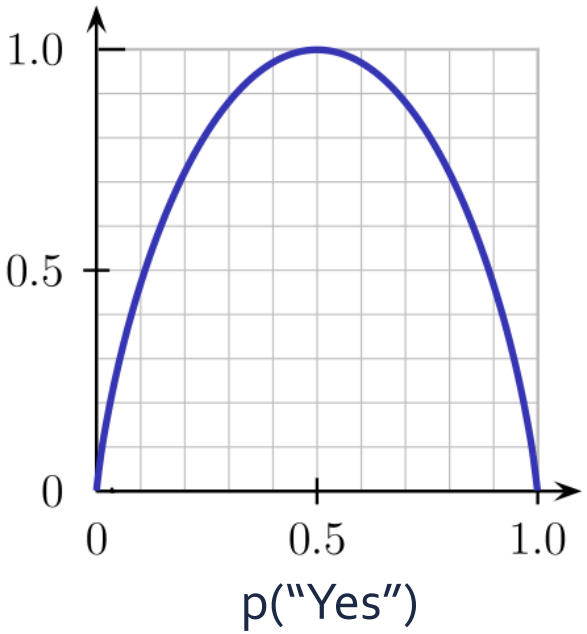


Entropy

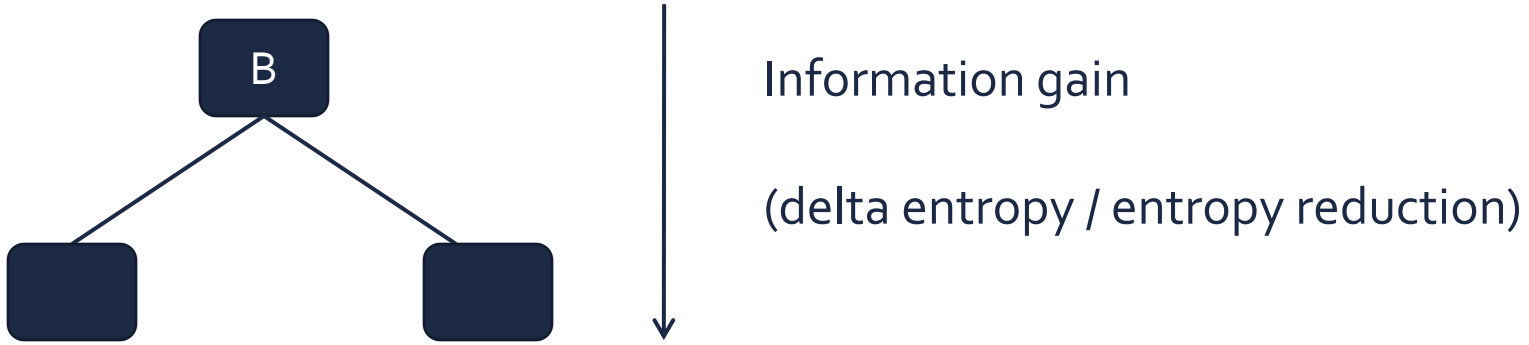
entropy(data_set) =

-1*(p(“Yes”)*$\log_2$(p(“Yes”))+ p(“No”)*log (p(“No”)))

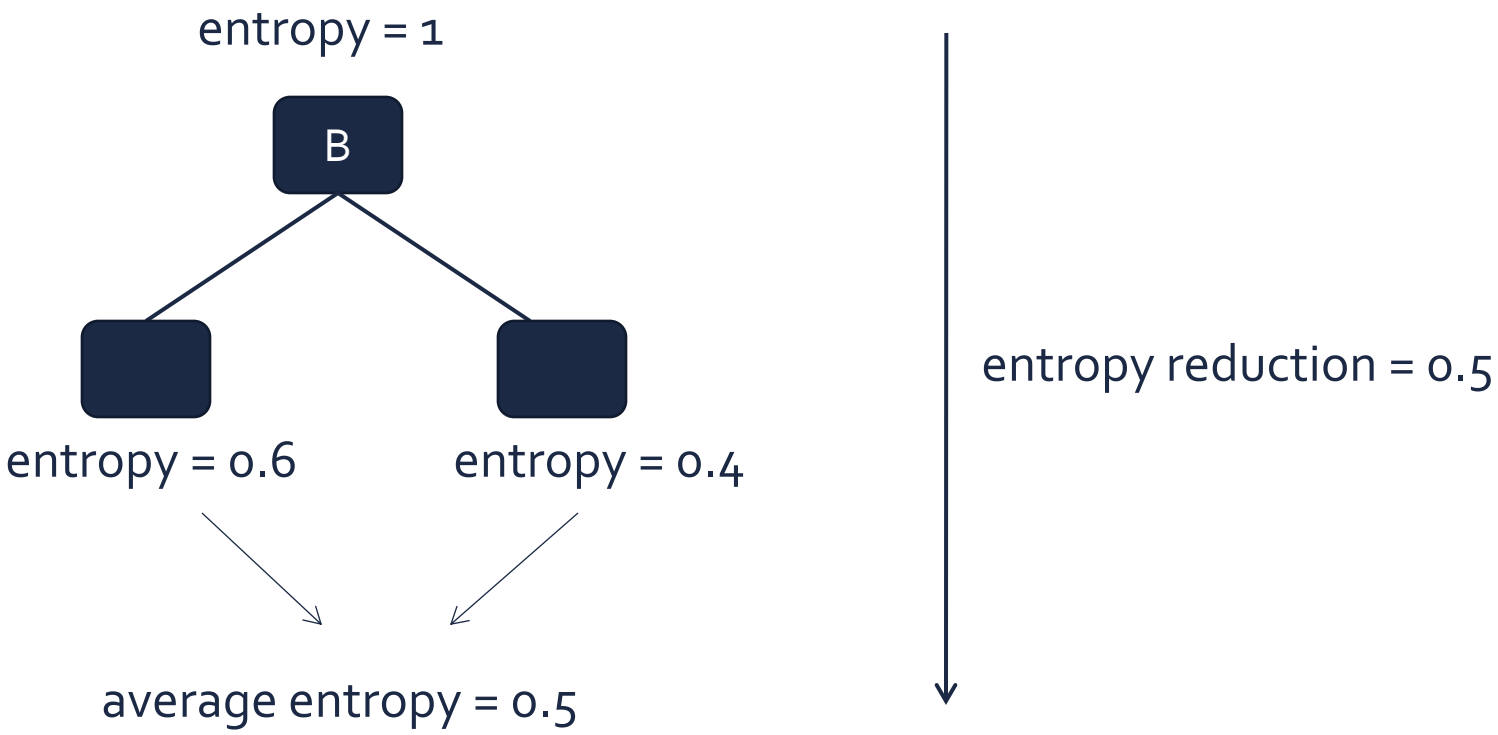
-1* ((13/26) $\log_2$(13/26) + (13/26) *log (13/26)) = 1



Information gain (delta entropy)



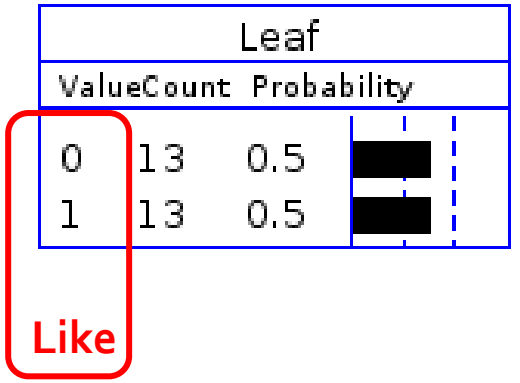
Information gain



Decision tree data structure

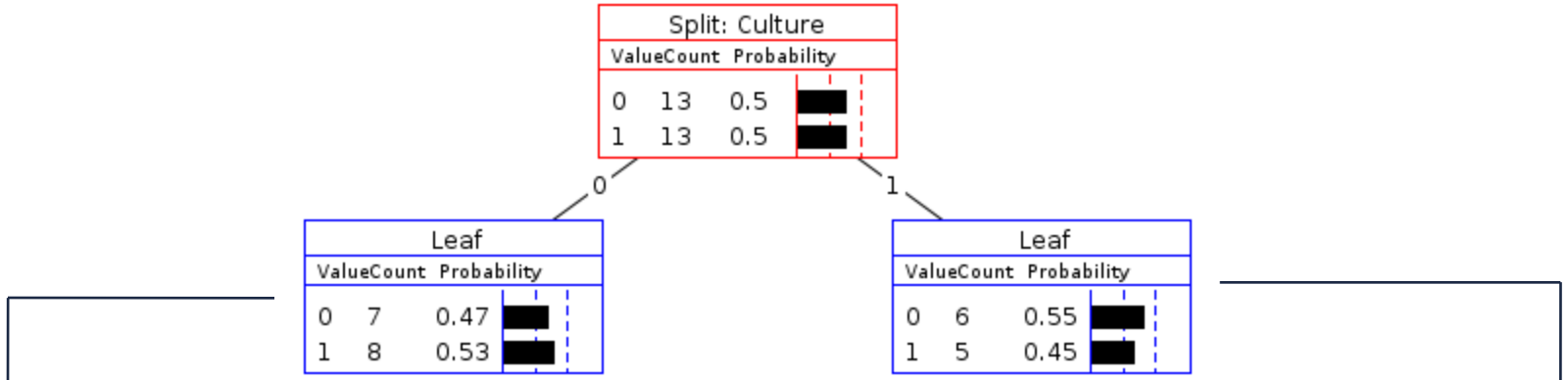
EXAMPLE STRUCTURE
[Numeric ID: Culture, Fly, Hot, Music, Nature, **Like**;]

Training examples:
01: 0, 0, 1, 0, 0, **0**;
02: 0, 1, 1, 0, 0, **0**;
03: 1, 1, 1, 1, 1, **0**;
04: 0, 1, 1, 1, 1, **0**;
05: 0, 1, 1, 0, 1, **0**;
…
26: 0, 1, 0, 1, 0, **1**;



Decision tree

entropy = 1

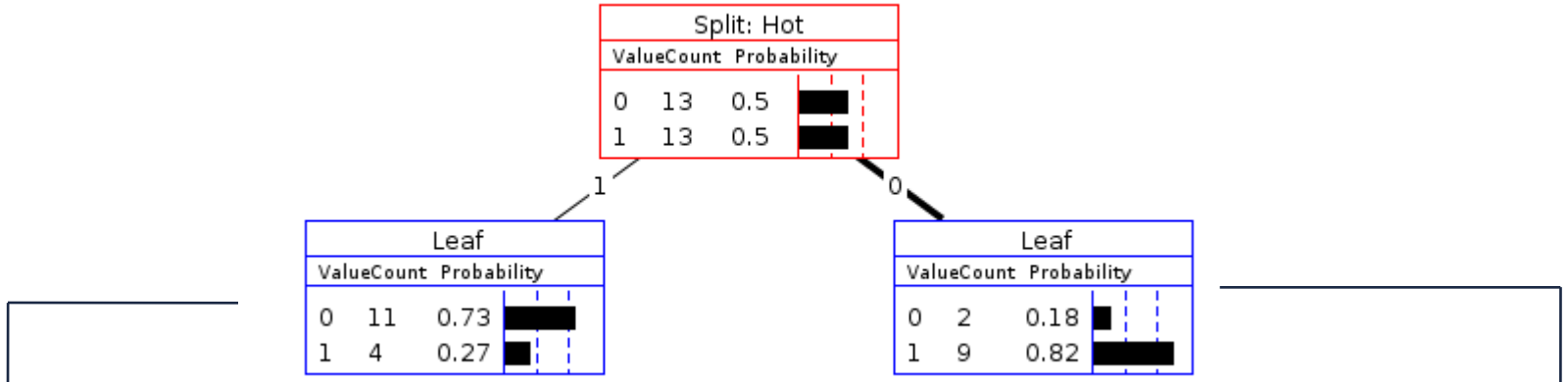


entropy = $-1 * (0.47*\log_2(0.47)+0.53*\log_2(0.53)) = 0.997$

entropy = $-1 * (0.55*\log_2(0.55)+0.45*\log_2(0.45)) = 0.991$

Decision tree

entropy = 1



entropy = -1 * (0.73*$\log_2$(0.73)+0.27*$\log_2$(0.27)) = 0.84

entropy = -1 * (0.18*$\log_2$(0.18)+0.82*$\log_2$(0.82)) = 0.68

Decision tree

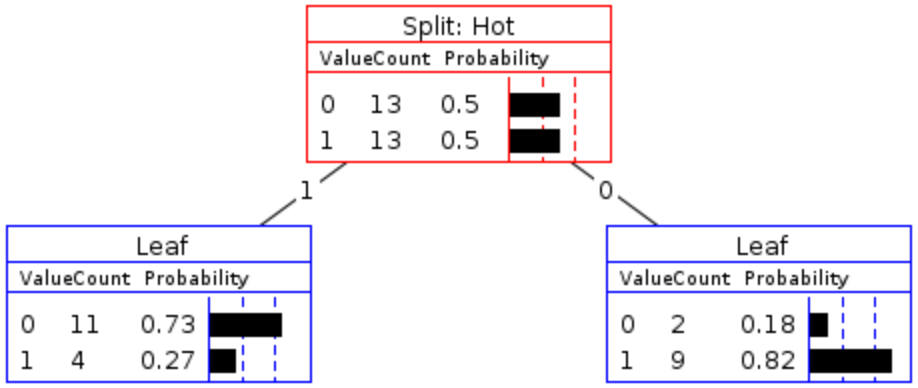
Attribute Name	Information Gain
☐ Culture	0.00
☐ Fly	0.00
☑ Hot	0.23
☐ Music	0.00
☐ Nature	0.16

Leaf

Value	Count	Probability
0	13	0,5
1	13	0,5

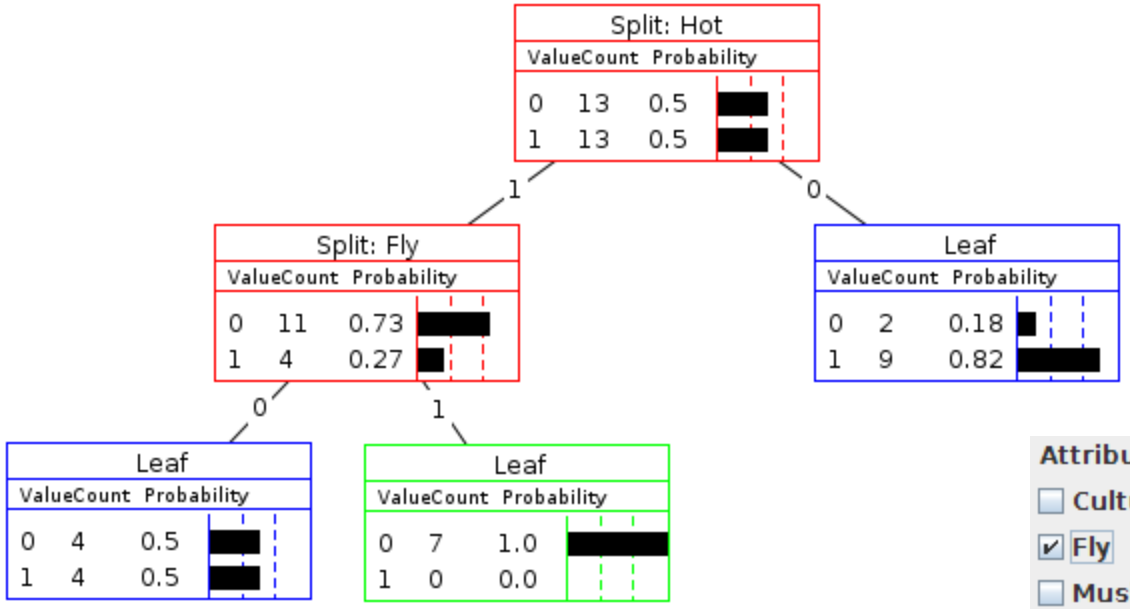


Decision tree



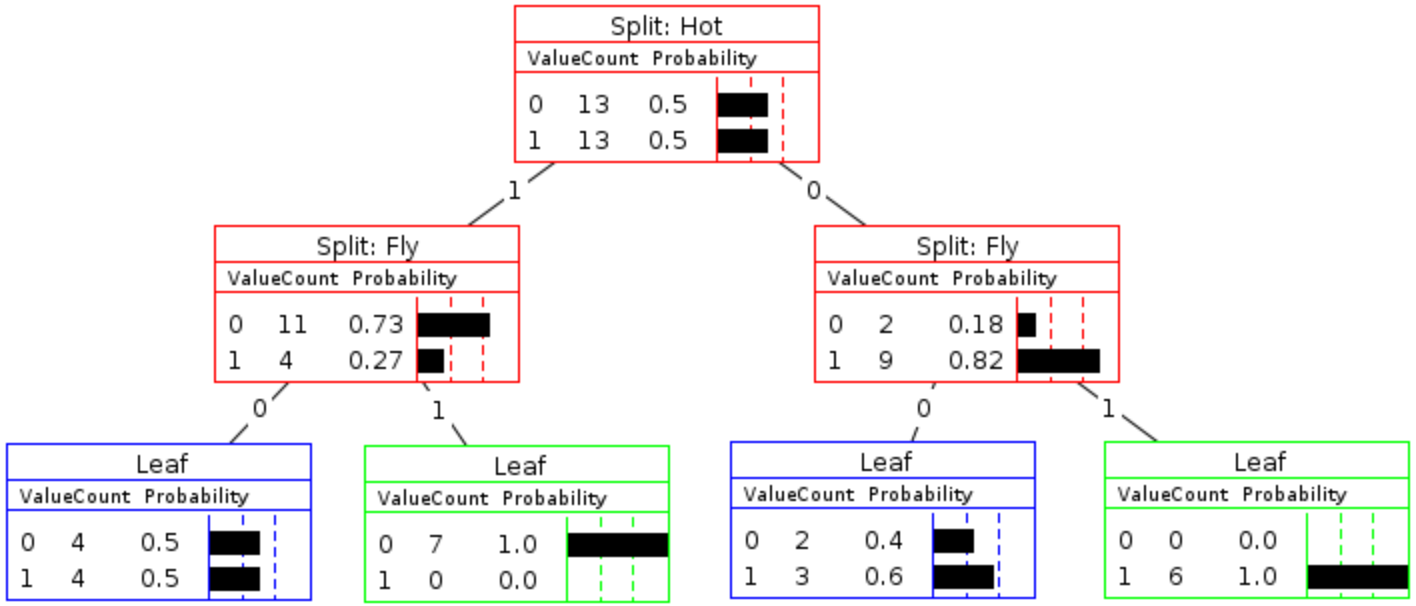
Attribute Name	Information Gain
☐ Culture	0.00
☑ Fly	0.30
☐ Music	0.00
☐ Nature	0.30

Decision tree



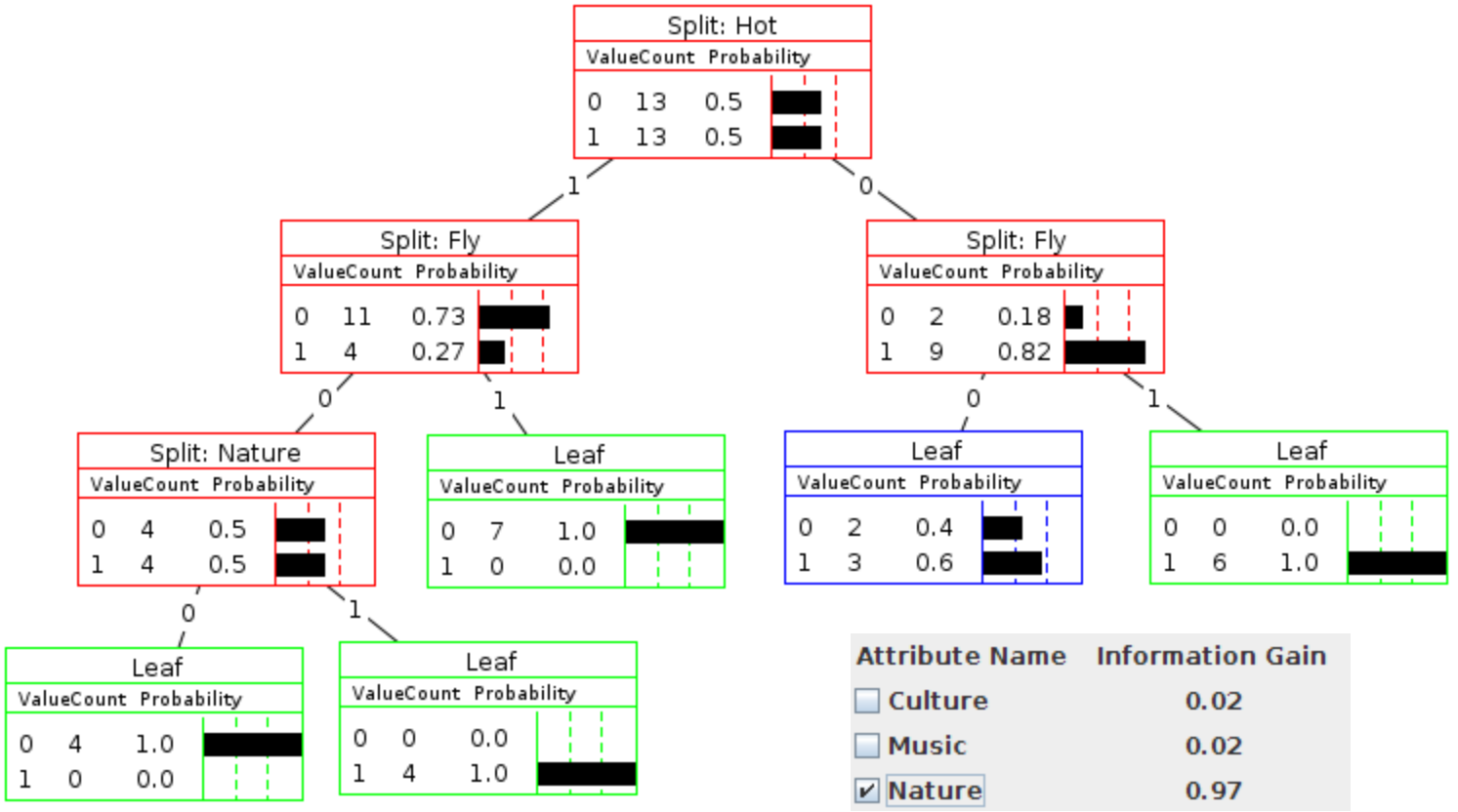
Attribute Name	Information Gain
☐ Culture	0.01
☑ Fly	0.24
☐ Music	0.00
☐ Nature	0.24

Decision tree



Attribute Name	Information Gain
☐ Culture	0.00
☐ Music	0.00
☑ Nature	1.00

Decision tree



Decision tree

